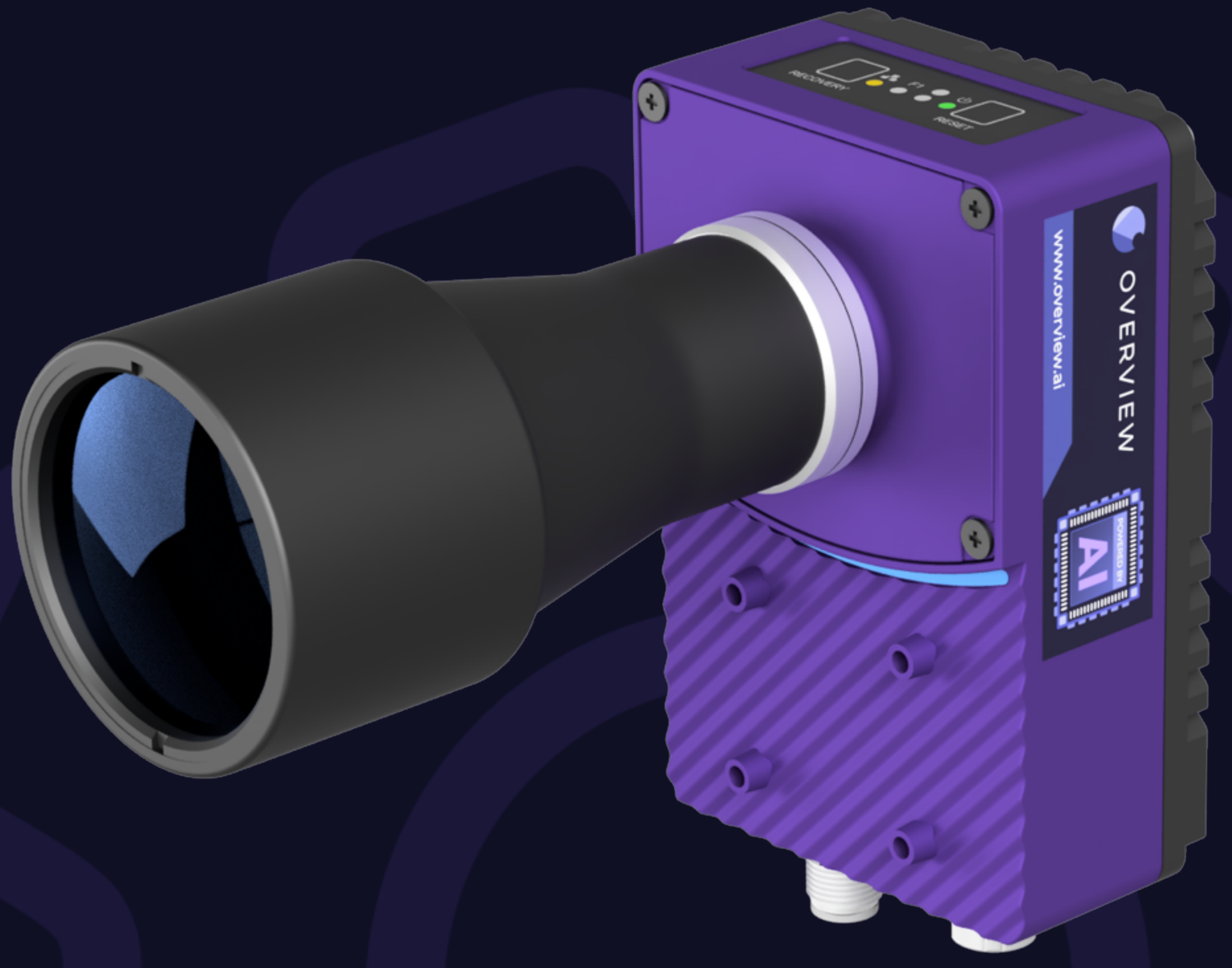
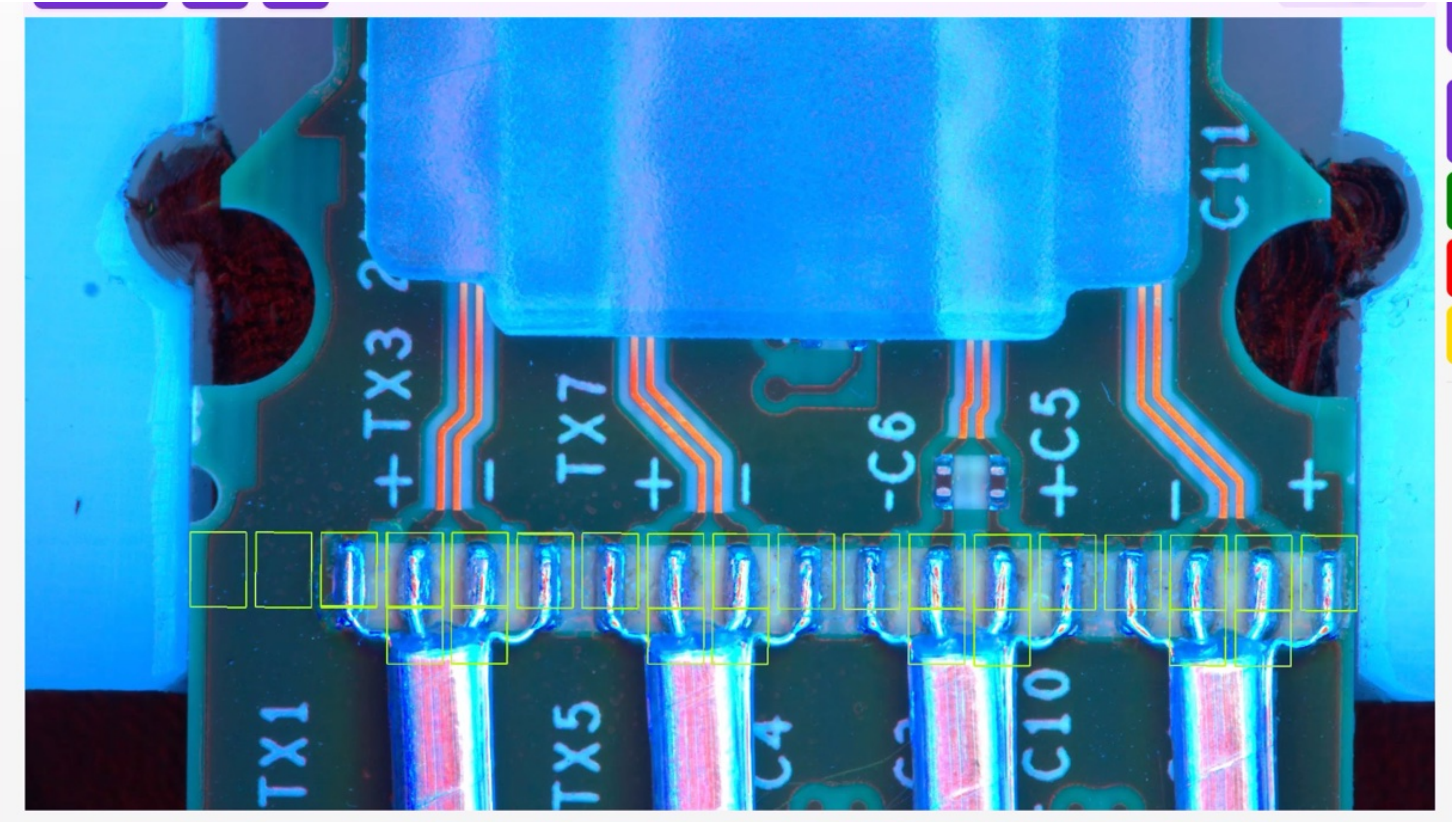
Hot Bar Soldering

OV80i Introduction



Hot Bar Soldering AI Inspection with OV80i

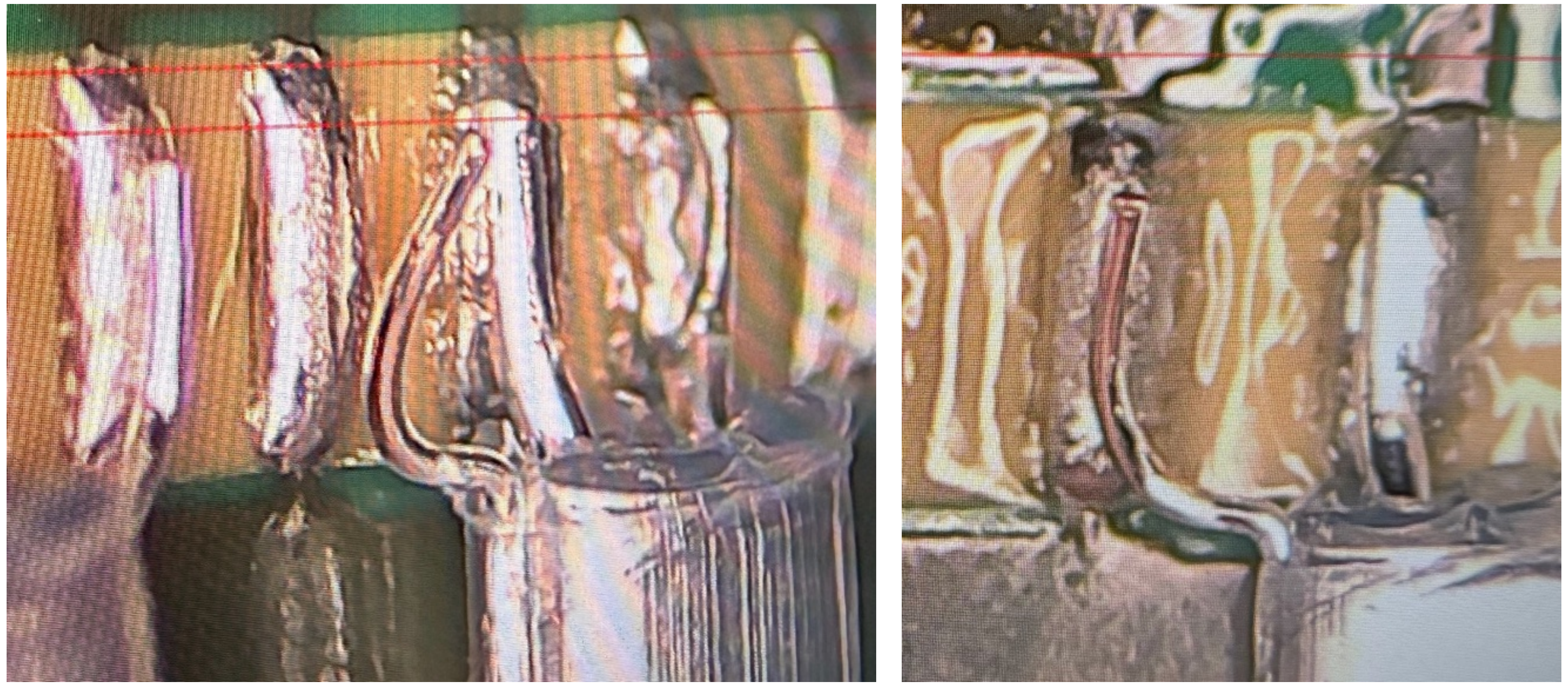
Successfully identified solder quality (including Z-axis defects), wire position, excess material.



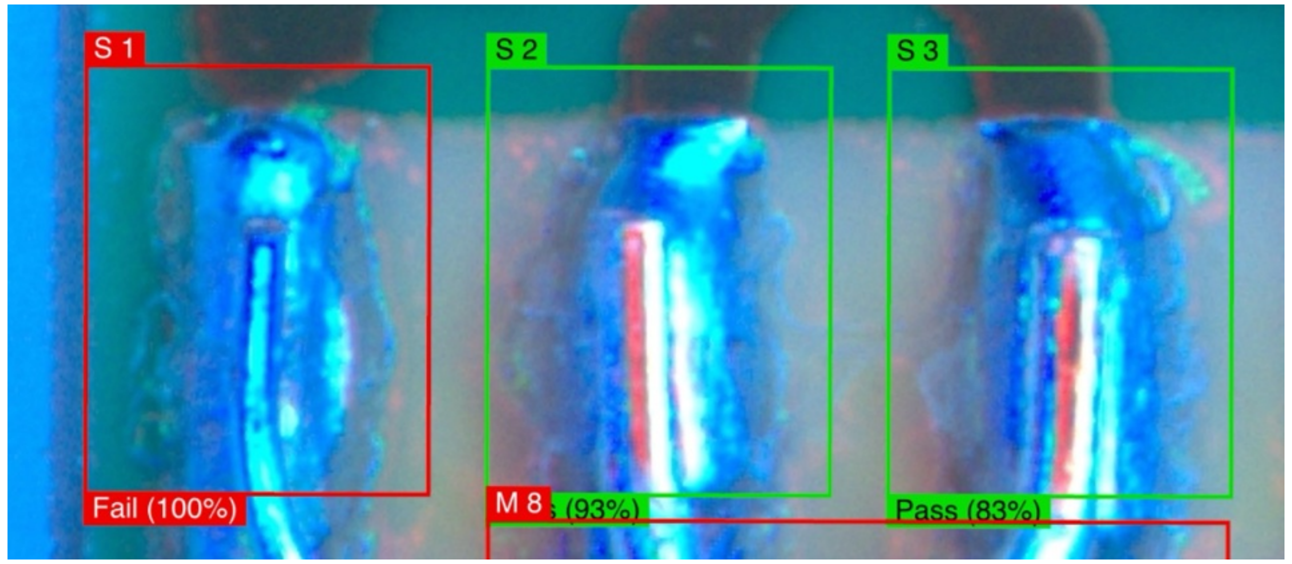
Catching Z-Axis Solder Defects

OV80i with advanced lighting and state-of-the-art AI catches most raised wire issues.

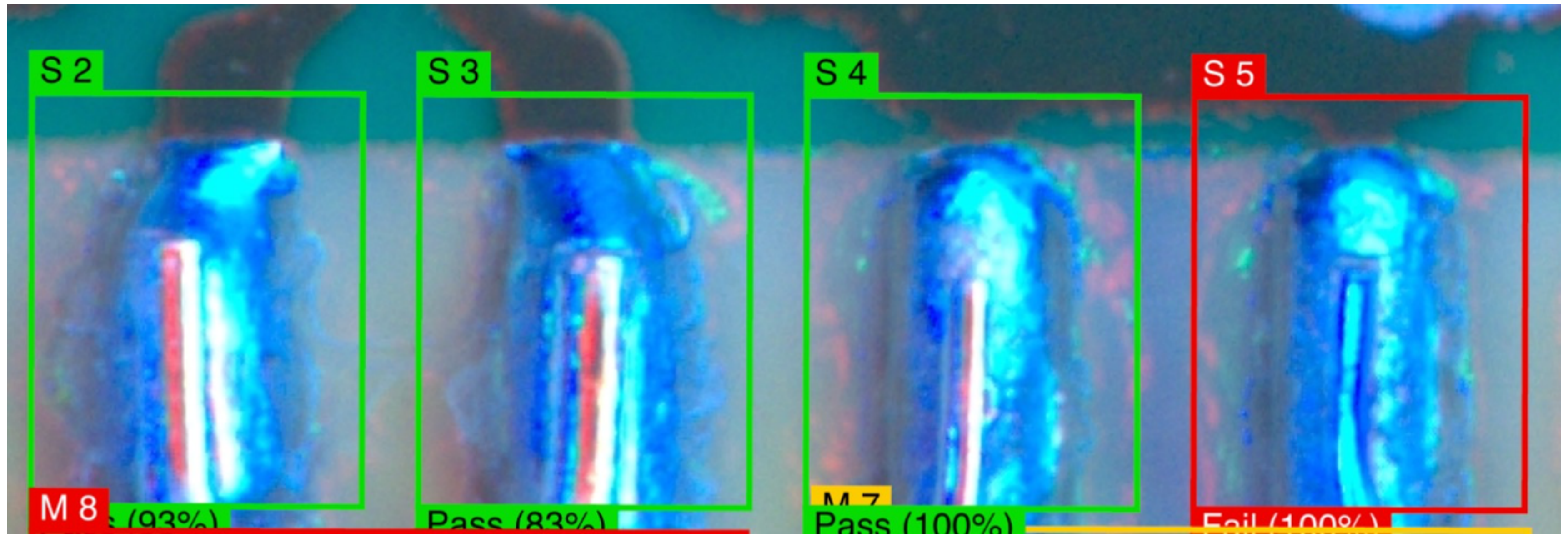
Very hard to see raised wire from top-down view (shown right)



Very hard to see raised wire from top-down view (shown right)

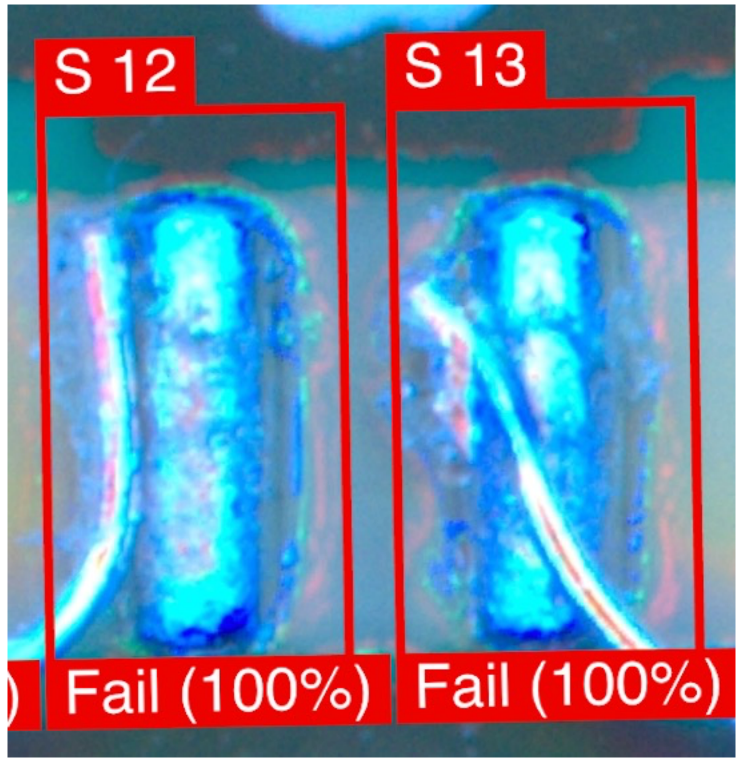


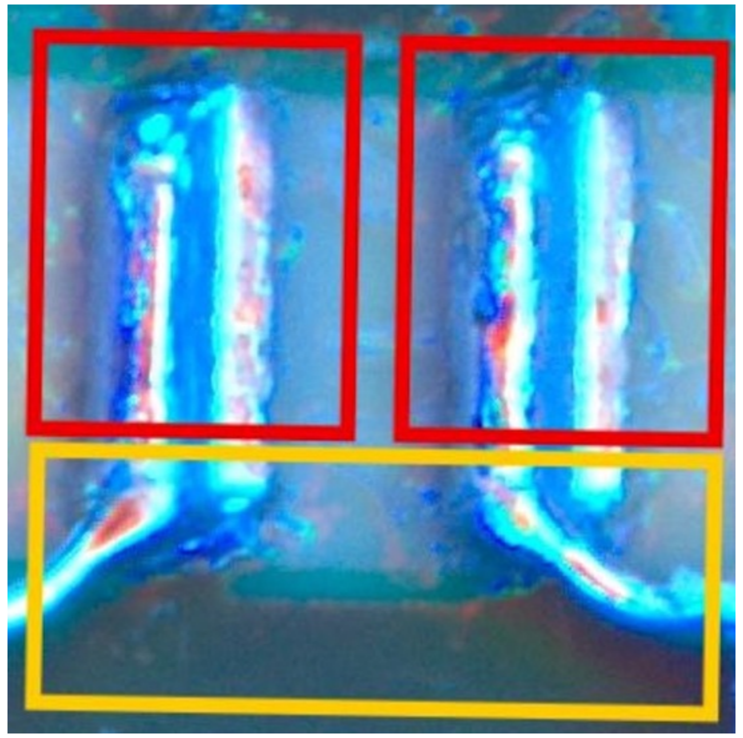
Special 2.5D lighting allows AI to defect this issue.

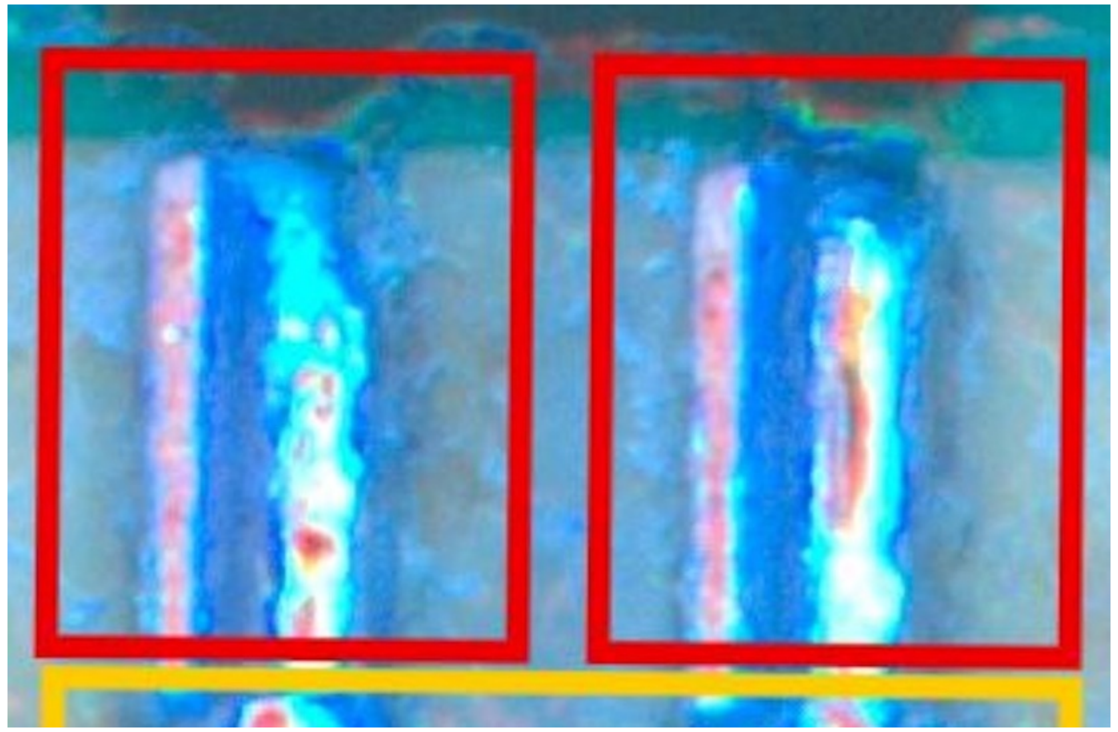


Special 2.5D lighting allows AI to defect this issue.

Example of Other Defect Types Caught







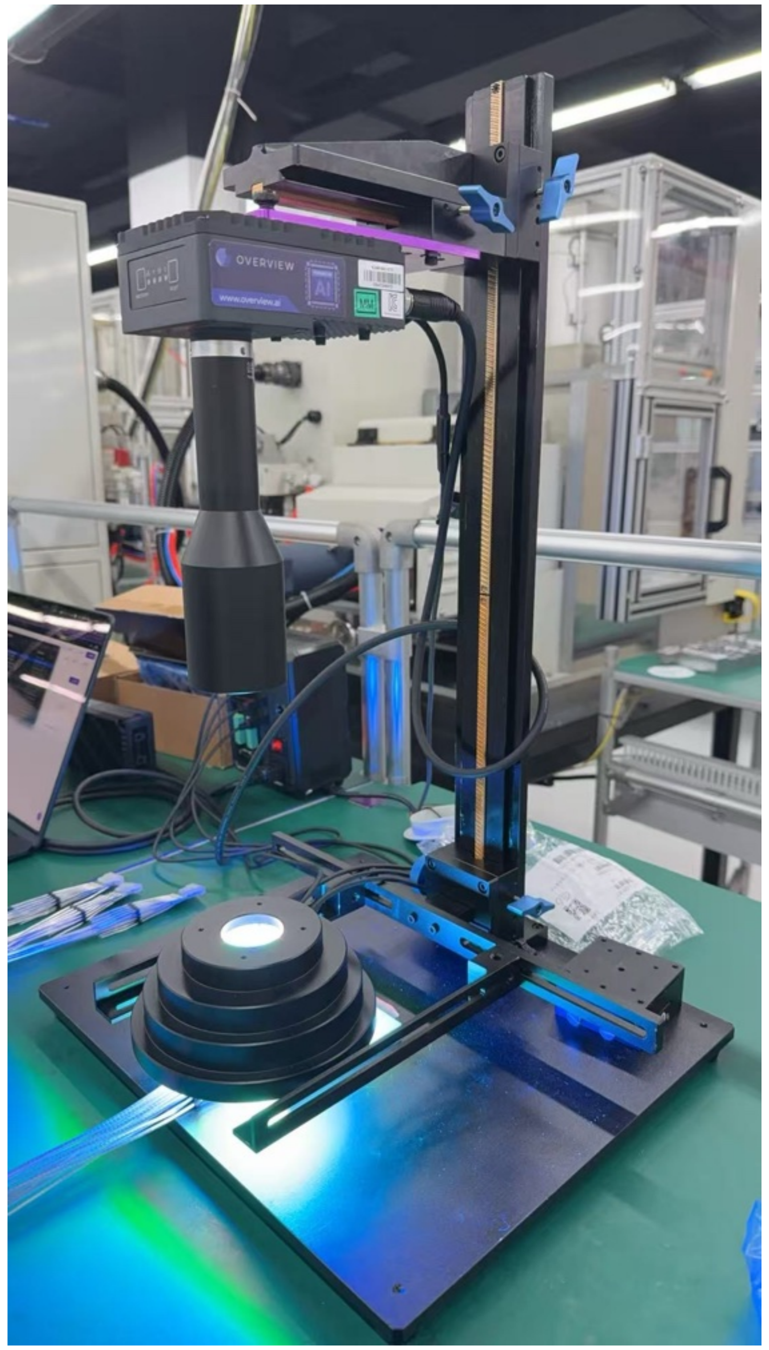
OVERVIEW

Configuring OV80i

Power of a vision system, ease-of-use of a vision sensor.

Step 1: Setup camera and lighting

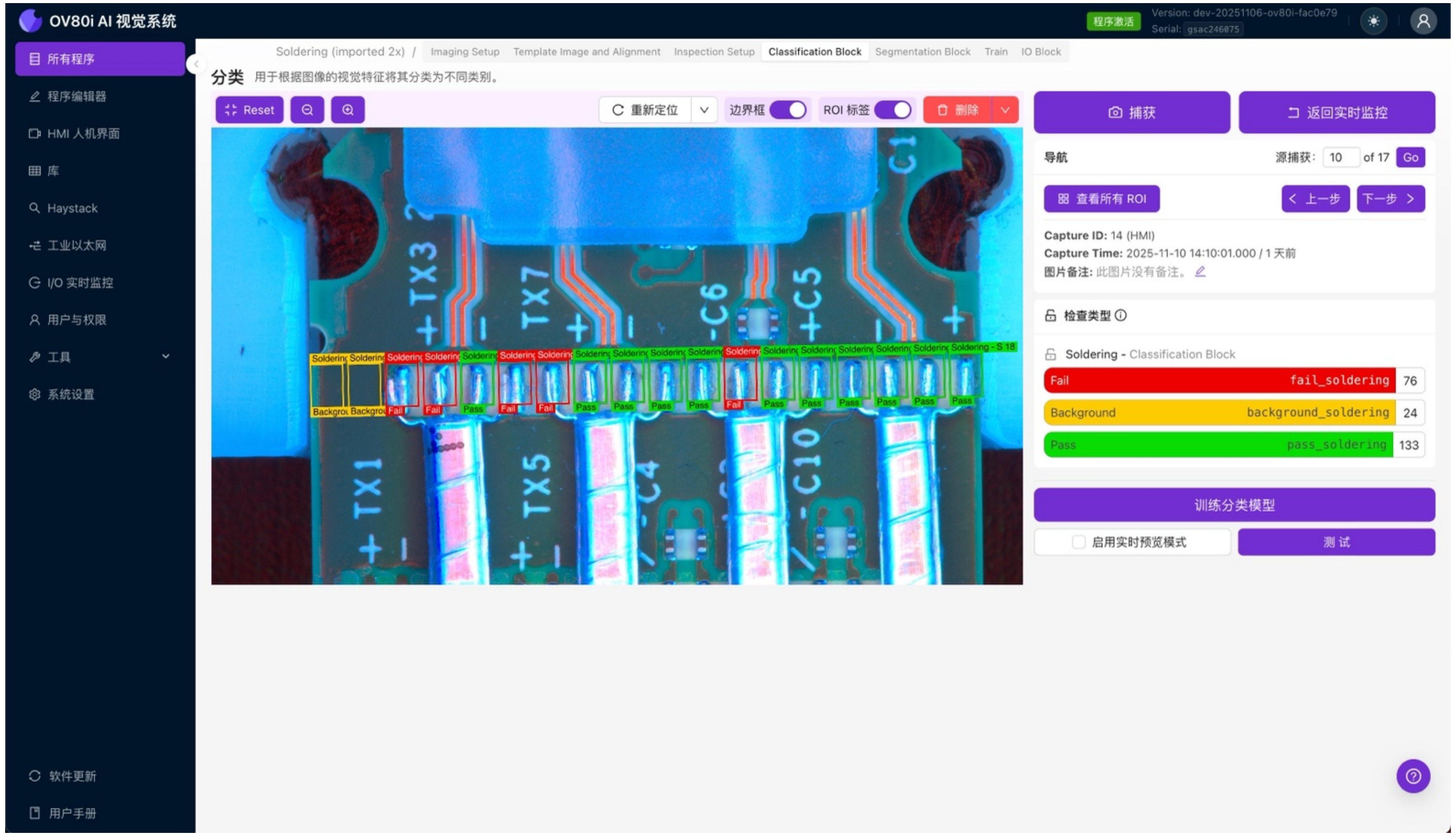
Optimizing image setup to increase accuracy and decrease difficulty of inspection.



1. **Telecentric Lens 0.4X Zoom for image consistency.**
2. **2.5D 3-Level Light to accentuate Z-axis issue (raised wire)**

Step 2: Setup classifier inspection

Add dynamic inspection regions per solder on PCB with auto-alignment tool. Only trained on 17 images!

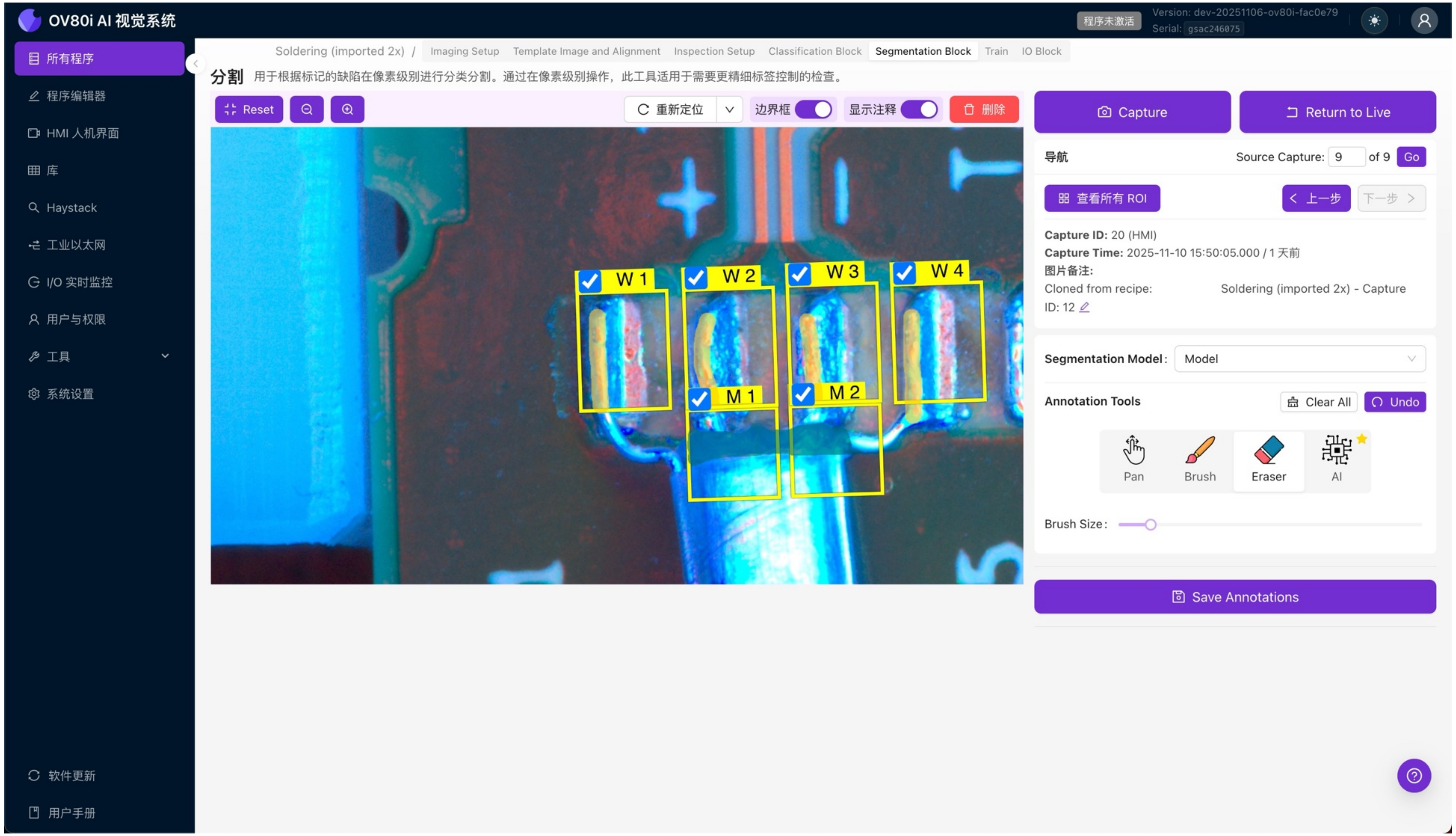


Easy setup: Simple labeling tools.

Multiple class options.

Step 3: Setup segmenter inspection

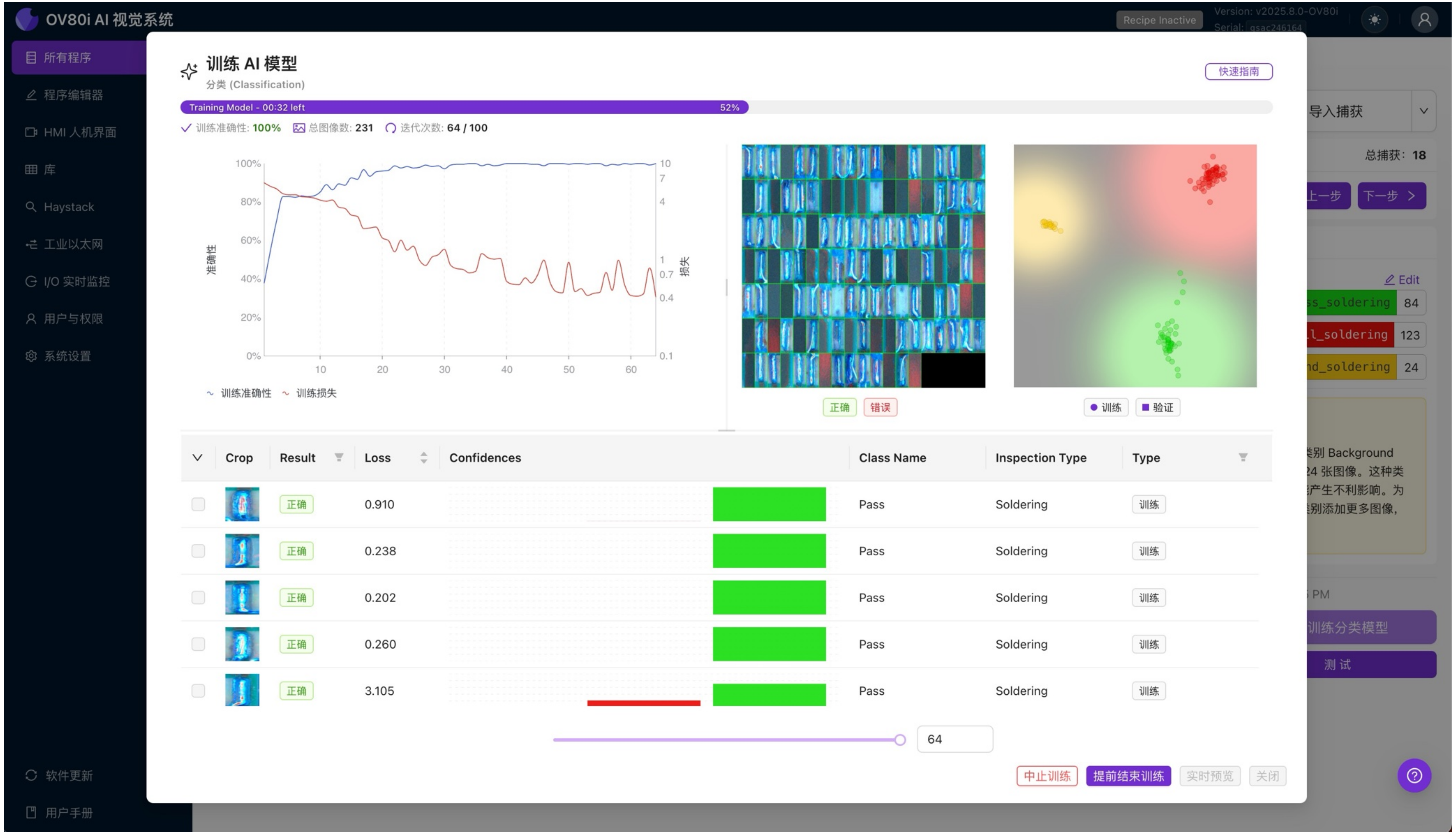
Layer on multi-class segmentation to find wire and check position + check for height and size of excess material. Only 9 images used!



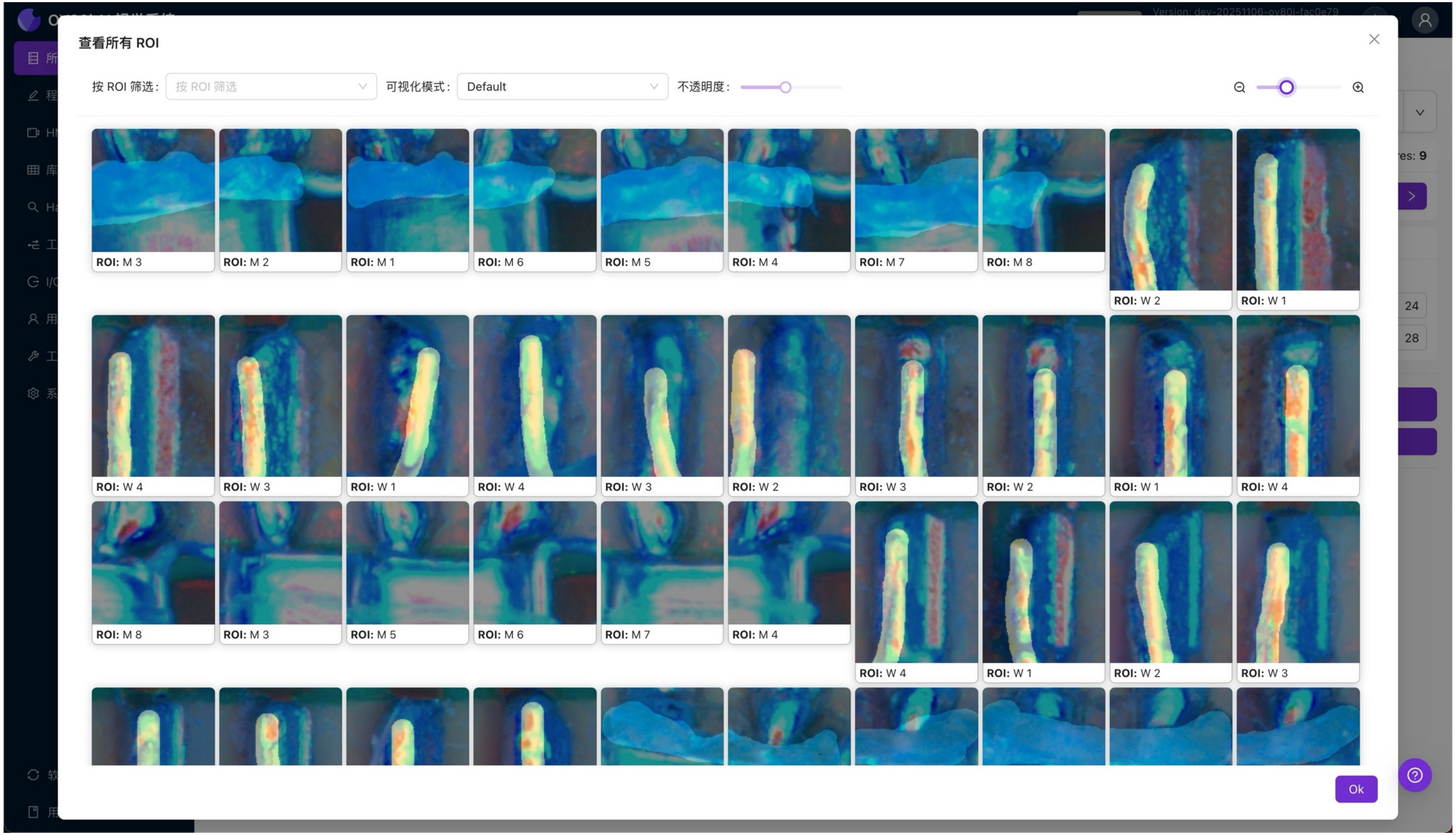
Step 4: Check classification dataset

Step 5: Train model and track mislabels

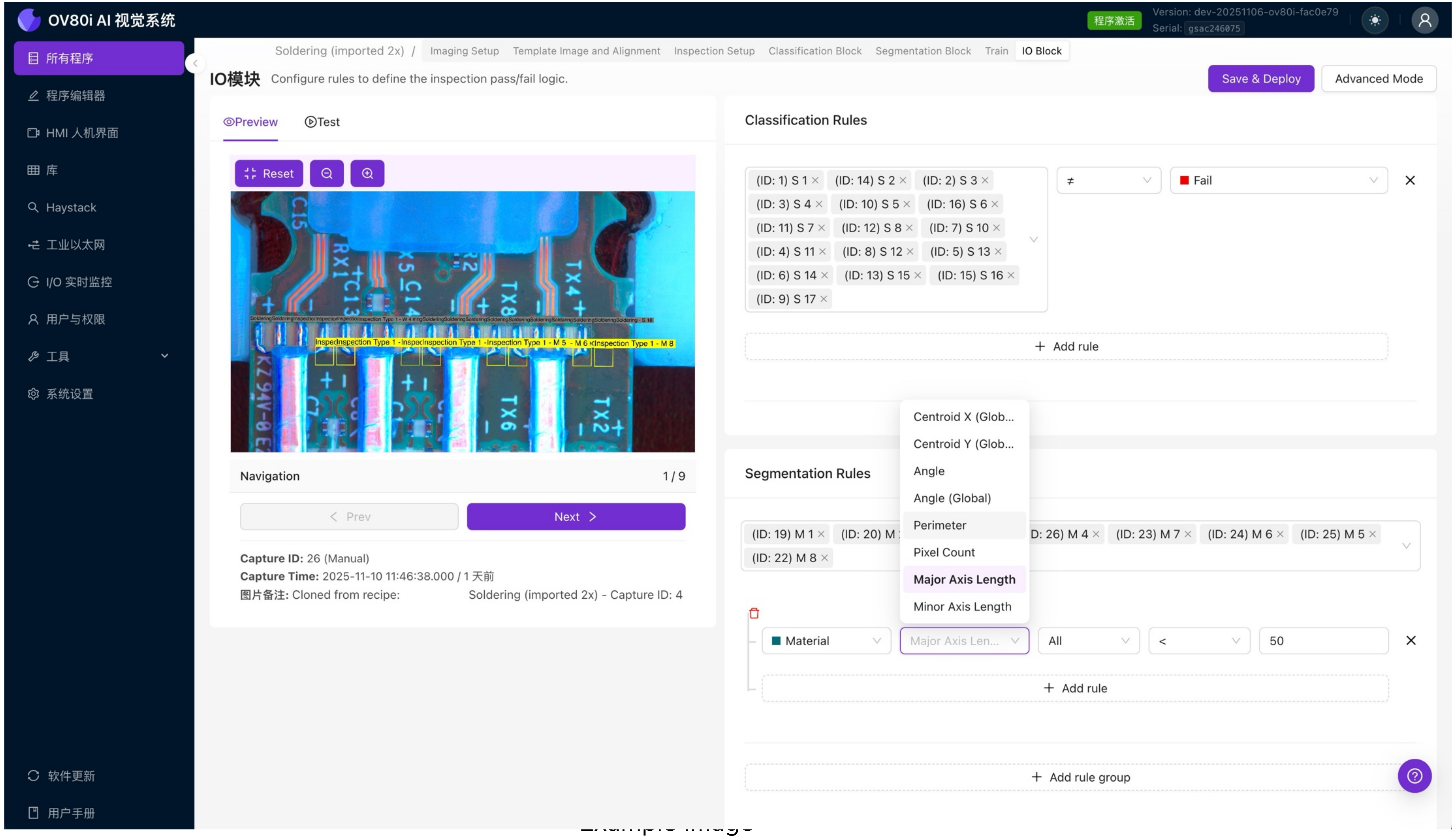
Training happens locally on OV80i in under 5 minutes.



Step 6: Check and train segmentation model



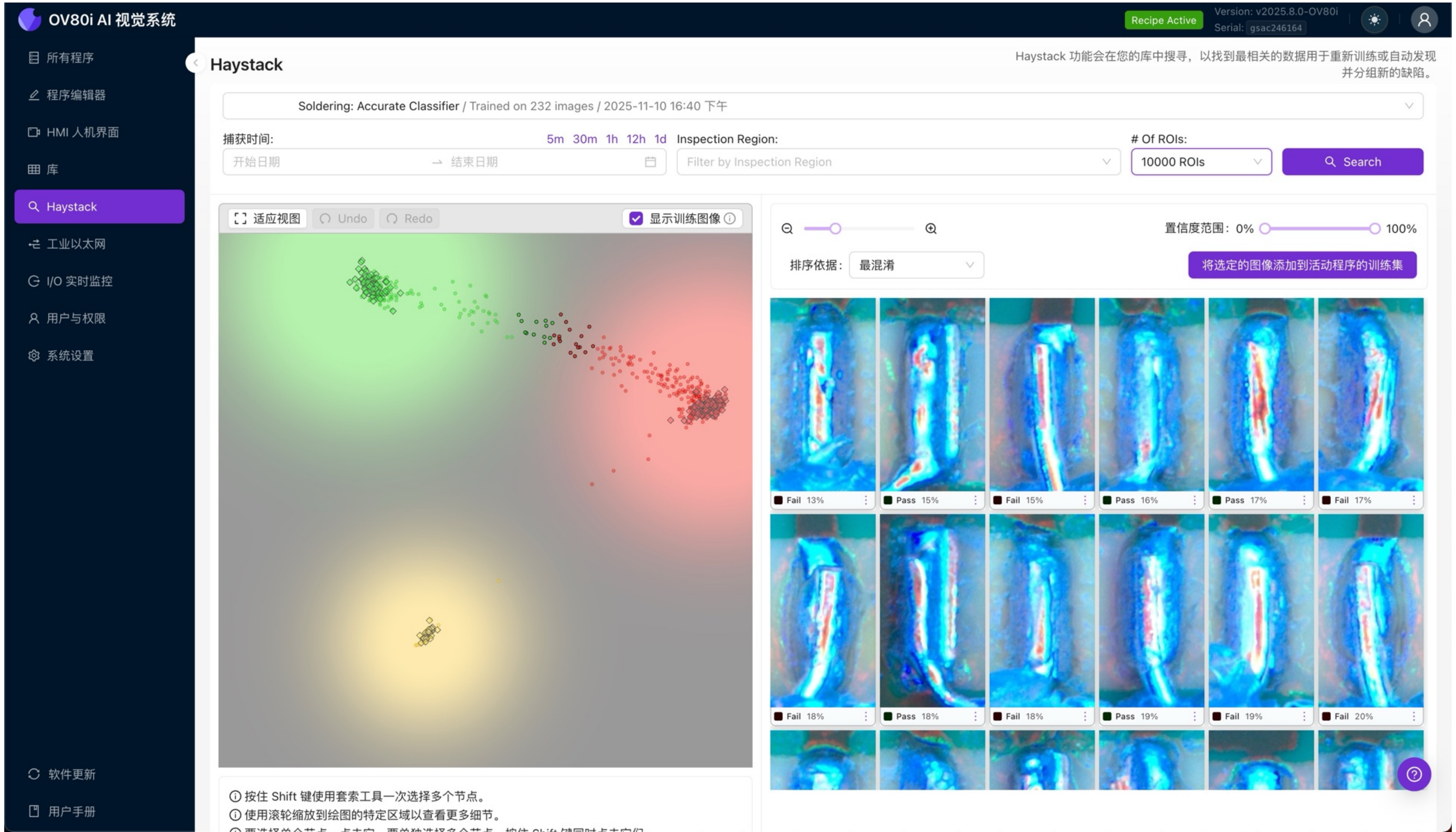
Step 7: Setup Data Output and Rule Builder



Results: 1-2 hours to configure, setup, label, train and deploy each inspection model.

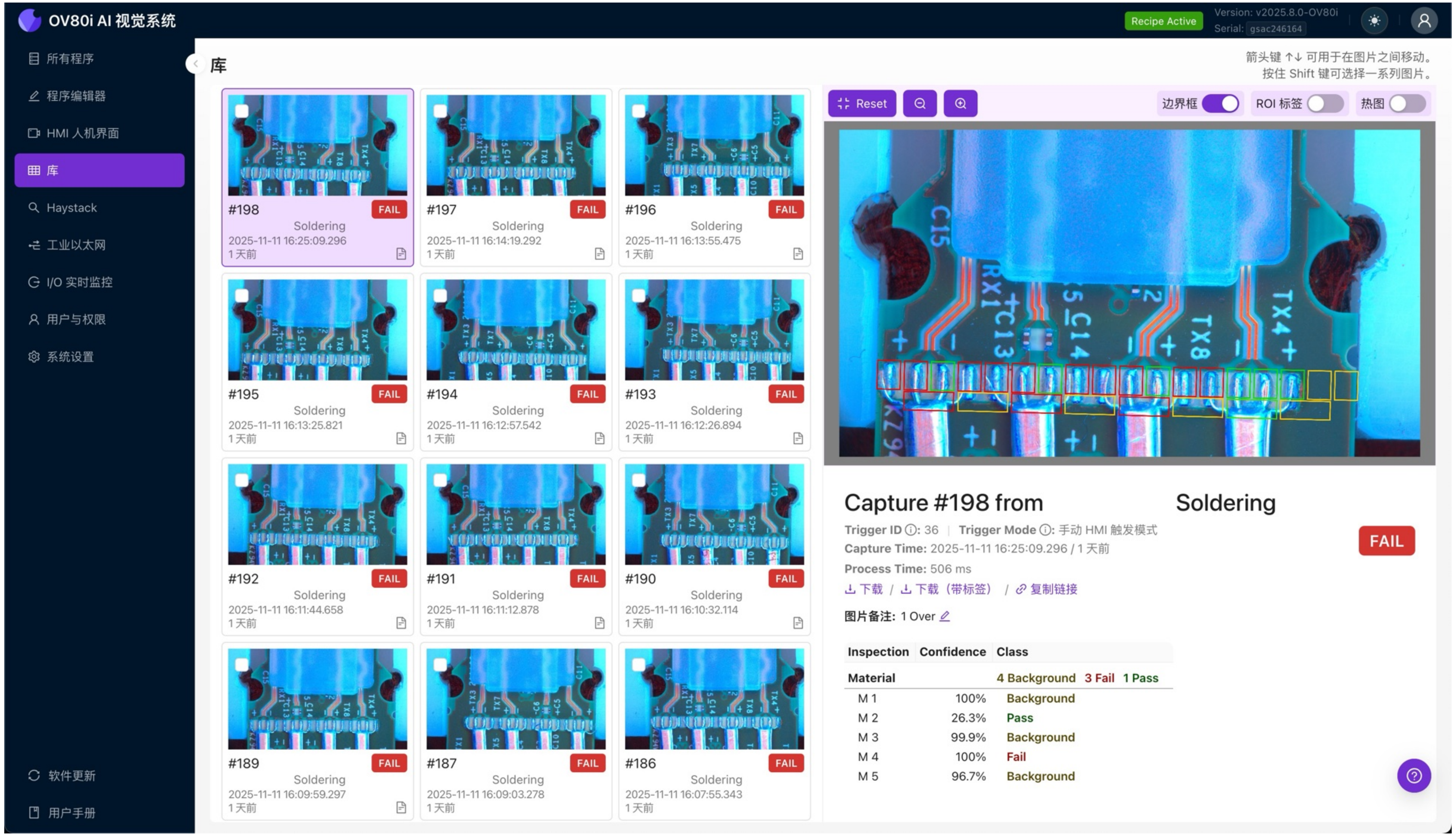
OV80i Haystack

Analyze thousands of production images stored on device. Look for outliers, borderline cases and find anomalies and/or new defect types.



OV80i Library: Store 20,000 images on device.

Easier root cause analysis, faster retraining and redeploying. Easy production data retraining.

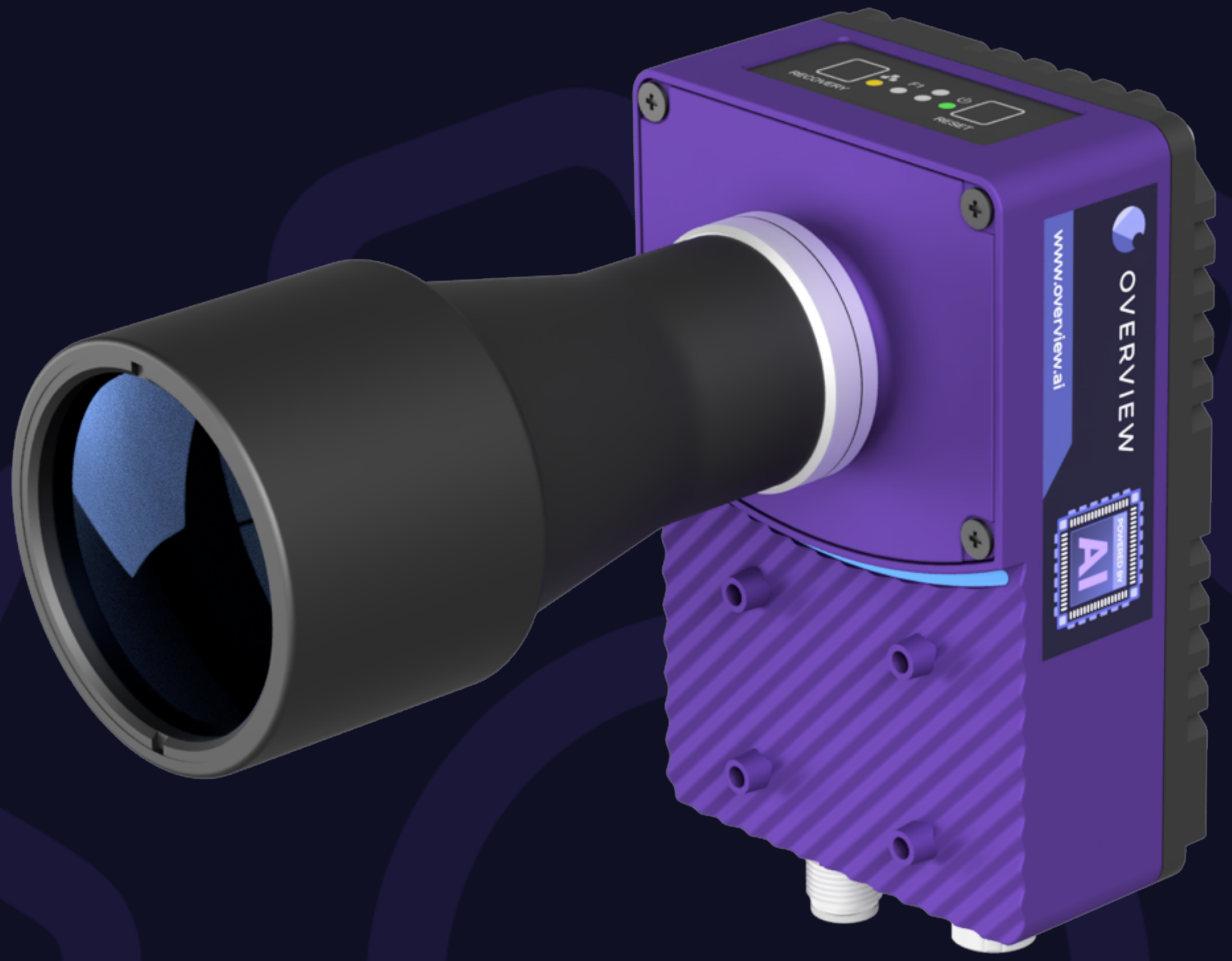


Custom notes stored in library by Engineer when double checking to verify inspection results.

OVERVIEW

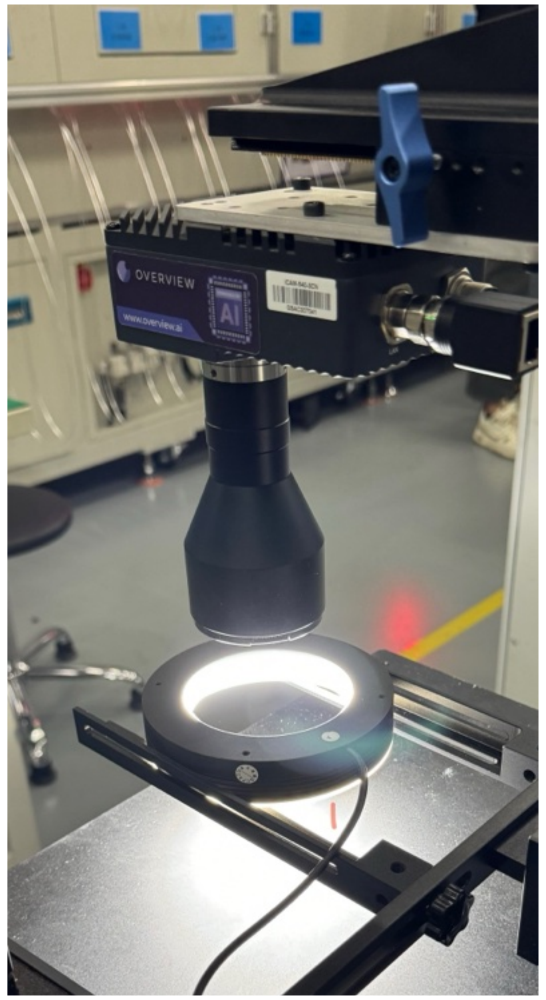
OV80i

AI Vision System



The OV80i

Designed for 10x Faster Installs. The truly all-in-one Deep Learning AI Vision System.



Imaging
8.3MP Color

On-board Library and Root-Cause Analysis Tool (Haystack)

Powerful Computing
- Onboard Nvidia GPU

No software download needed

Industry Leading Accuracy. Built on reliable, proven hardware. No software downloads or special computer requirements. *Easy setup*, fast ramp-up for engineers and technicians. Rapid troubleshooting and maintenance.

No software licenses, no recurring fees.

OV80i Benefits and Advantages

1. Industry leading AI accuracy. Form-factor and ease-of-use of vision sensor.
2. Reduced variance in manual human inspection, AI inspection is 100% inspection, 24/7.
3. Reduce escapes of bad parts.
4. Onboard data. Simplify root cause analysis and eventual yield gains.
5. Easy to train – AI models can be trained in 30 minutes or less by manufacturing, process, controls or quality engineers.
6. Flexibility – OV80i can be used for small component defect detection or larger assembly verification tasks.
7. Fast inspections – as low as 0.3 seconds per image.
8. Visual AI Inspections that can help replace human inspectors.
9. Multi-language Support.

OVERVIEW

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